

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final
 Duration: 2 hours

Semester: Summer 2024
 Full Marks: 50

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
 Figures in the right margin indicate marks.

SET A

Name:	ID:	Section:
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1. [CO2]	a)	Consider a multi-level ASK where 2 bits are sent in each baud. The carrier signal has 2 full-cycles in each signal element, and a phase of 0°. Amplitude for each 2-bits pattern is given as: 00 $\rightarrow$ 5V, 01 $\rightarrow$ 10V, 10 $\rightarrow$ 15V, 11 $\rightarrow$ 20V. Based on this, draw the modulated signal in the following graph for the bit sequence 001110011101.	[5]
	b)	For the QPSK-modulated signal given below, draw the constellation diagram, and find the amplitude and phase of the modulated signal.	[2+1+2]

2. [CO3]	a)	The minimum bandwidth required to multiplex 6 channels is 555 kHz. There are 15 kHz guard bands used to prevent interference for each channel. Calculate the bandwidth of each input channel. Show the required bandwidth with a diagram where the starting frequency is 101 kHz. What will be the highest frequency?	[2+2+1]																	
	b)	Mention what advantage does statistical TDM provide over synchronous TDM.	[2]																	
	c)	In a network company, where multiple departments share a dedicated communication link for their daily operations, the network administrator allocates a fixed time slot to each department regardless of the actual data demand at any given time. Based on this scenario, identify which TDM is used and why?	[3]																	
	d)	Consider 3 channels, each having different input data rates. If the data rate of the first channel is 12 kbps, then the rest of the channels have double the data rate of their previous channel. You need to multiplex them in such a way, so that the number of channels increases. The interleaved unit is 4 bits along with 1 bit for synchronization. Calculate the following: i. Draw the figure to visualize the multiplexing process with proper labels ii. Input bit duration iii. Frame Size iv. Frame Rate v. Output bit duration	[5]																	
3. [CO4]	a)	The following codeword is received by a receiver X. Receiver X uses CRC technology to detect errors. Determine whether there is any error in the received codeword or not with proper calculation. [The first 2 bits in the divisor are 1, rests are 0] Codeword <table border="1"><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr></table> <table border="1"><tr><td colspan="5">Dataword</td></tr></table> <table border="1"><tr><td colspan="3">Remainder</td></tr></table>	1	0	0	1	1	0	1	0	0	Dataword					Remainder			[5]
1	0	0	1	1	0	1	0	0												
Dataword																				
Remainder																				
	b)	Suppose we are using a block coding scheme with a (3,5) code. How many Dataword can be encoded ? How many Codewords will be invalid ?	[1+1]																	

	c)	Suppose we are using a (2,4) encoding scheme and the assignment of Dataword to Codewords are as follows: <table><tr><th><u>Dataword</u></th><th><u>Codeword</u></th></tr><tr><td>00</td><td>0010</td></tr><tr><td>01</td><td>1001</td></tr><tr><td>10</td><td>1011</td></tr><tr><td>11</td><td>1100</td></tr></table> i. What is the minimum Hamming distance here? ii. How many bits can be corrected and detected? iii. The receiver receives 0111. Explain what type of error this is. Can this codeword be corrected? Justify your answer.	<u>Dataword</u>	<u>Codeword</u>	00	0010	01	1001	10	1011	11	1100	[2+2 +1+3]
<u>Dataword</u>	<u>Codeword</u>												
00	0010												
01	1001												
10	1011												
11	1100												
4. [CO4]	a)	There are three stations sharing a communication link. Station 1 is on the left side, Station 3 is on the right side, and Station 2 is in between. Assume they are using CSMA/CD. • Station 1 starts sending a message to Station 3 at time t = 4 ms • The first bit of the message reaches Station 2 at time t = 12.4 ms and Station 3 at time t = 16.1 ms • Station 1 finishes sending the message at time t = 20.4 ms Based on the given information, calculate the followings: i. How long does it take Station 1 to send the message (Frame Transmission time)? ii. What is the longest time it takes the message to travel between stations (Maximum Propagation time)? iii. If the link speed is 2 Gbps, what is the Minimum Frame Size required to meet the conditions given?	[2+2 +3]										
	c)	Why is the vulnerable time in slotted ALOHA halved of the pure ALOHA?	[3]										

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